UNIVERSITY OF CALIFORNIA,
IRVINE

A High and Variable Dimensional Measurement of the Z+jets Differential Cross Section with the ATLAS Experiment and Aritifical Intelligence

DISSERTATION

submitted in partial satisfaction of the requirements
for the degree of

DOCTOR OF PHILOSOPHY

in Physics and Astronomy

by

Kevin Greif

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Daniel Whiteson, Chair
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2026

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DEDICATION

(Optional dedication page)

To Dario Amodei mostly I guess?

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ACKNOWLEDGMENTS

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You may also acknowledge the contributions of professors and friends.

You also need to acknowledge any publishers of your previous work who have given you permission to incorporate that work into your dissertation. See Section 3.2 of the UCI Thesis and Dissertation Manual.)

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1. V. Breso-Pla et al.. "Explicit or Implicit? Encoding Physics at the Precision Frontier". *Preprint:* arXiv:2603.08802, 2026.

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3. A. Shmakov et al.. "Full event particle-level unfolding with variable-length latent variational diffusion". *SciPost Phys.* 18(4):117, 2025.

4. N. Huetsch et al.. "The landscape of unfolding with machine learning". *SciPost Phys.* 18(2):070, 2025.

5. ATLAS Collaboration. "Accuracy versus precision in boosted top tagging with the ATLAS detector". *JINST* 19(08):P08018, 2024.

6. A. Shmakov et al.. "End-to-end latent variational diffusion models for inverse problems in high energy physics". *NeurIPS Proceedings*, 2023.

7. K. Greif and K. Lannon. "Physics Inspired Deep Neural Networks for Top Quark Reconstruction". *EPJ Web Conf.* 245:6029, 2020.

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Nov. 2023	"End-to-End Latent Variational Diffusion Models for Unfolding LHC Events", *ML4Jets 2023*
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- Physics 7LC (Mechanics Lab), Winter 2021
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- Computational Methods in Physics, Spring 2020

OUTREACH

ATLAS Virtual Visit Tour Guide **2022–2025**
Remote visits to the ATLAS experiment for students from around the world.

UCI Physics and Astronomy Blog **2020–2022**
Published accessible descriptions of new research from the UCI physics department.

ABSTRACT OF THE DISSERTATION

A High and Variable Dimensional Measurement of the Z+jets Differential Cross Section with the ATLAS Experiment and Aritifical Intelligence

By

Kevin Greif

Doctor of Philosophy in Physics and Astronomy

University of California, Irvine, 2026

Daniel Whiteson, Chair

Proton-proton collisions at the Large Hadron Collider (LHC) offer the opportunity to observe the interactions of fundamental particles at very high energy scales. The high collision rate and large number of final state particles produced per collision imply that the datasets produced by detectors such as the ATLAS experiment are large and highly dimensional. The complexity of these datasets calls for the use of novel data analysis techniques which can exploit all of the available information to illuminate known interactions and search for new ones. This thesis presents applications of artifical intelligence (AI) techniques to improve the physics results of the ATLAS experiment. It centers on a full phase measurement of the Z+jets production cross section at the LHC, where the cross section is measured differential in the kinematics of every final state charged particle through the use of an AI based unfolding algorithm. This is the first such measurement performed at the LHC, which provides a complete experimental characterisation of the Z+jets production process and a proof-of-principle for the further pursuit of such measurements on a host of LHC processes.

Chapter 1

Introduction

The discovery of the Higgs Boson, announced on July 4th, 2012 [1, 2], is widely regarded as the most important discovery in particle physics in the 21st century. With that discovery the electroweak symmetry breaking mechanism [3–5] received experimental confirmation and the Standard Model of particle physics [6, 7] was complete. This triumph of modern science was rightly front-page news, and is still well known in the popular imagination. A less well known, though equally important, advancement was announced 87 days later on September 30th, 2012 when the results of the "ImageNet Large Scale Visual Recognition Challenge" were released. This was a challenge within the computer vision research community to develop a machine learning model that could sort images into 1000 classes. The training set for this challenge consisted of 1.2 million painstakingly hand-labeled images, which was unprecedented scale for labeled datasets at the time. This scale and the recent release of Nvidia's CUDA toolkit, which allowed for optimized matrix operations on GPU hardware, allowed for a team of researchers from the University of Toronto to train a deep convolutional neural network named AlexNet that outperformed its closest competitor by 10% in top-5 error rate[1] [8]. Before this moment state-of-the-art computer vision models relied on hand-

[1]The top-5 error rate is the percentage of images that were not correctly classified into the top 5 most likely classes.

engineered features programmed by human experts, but AlexNet learned all features used to classify the images directly from the data, requiring the humans who programmed it to know essentially nothing about the images themselves. This breakthrough became known as the "AlexNet moment" and marked the beginning of the modern era of machine learning.

This thesis is an attempt to understand the impact of the AlexNet moment on particle physics research in the post-Higgs era. It's worth dwelling very briefly on the fact that both discoveries lead to a Nobel prize in physics, the Higgs for Peter Higgs and François Englert in 2013 and AlexNet for Geoffrey Hinton in 2024. The nobel prize for Higgs and Englert was, by my count, the 12th Nobel prize awarded for the development of the Standard Model[2]. It was the capstone of a very fruitful and now very mature research program. By contrast the Nobel for AlexNet was at least in-part an "instrumentation" award[3], significant for the other discoveries it enables. Particle physics was one of the first scientific disciplines to make use of deep learning. The first papers demonstrating the use of deep learning for event selection [9] and jet tagging [10] were written in 2014, only two years after AlexNet. Since then there has been a remarkable growth in the number of papers applying deep learning to particle physics [11]. This thesis will provide a tour of some of these applications. Concurrently, most other sub-fields of physics have taken interest in applying deep learning tools. Cosmology, astrophysics, gravitational wave science, condensed matter physics, plasma physics, fluid dynamics, quantum information, and optical physics have all seen fruitful applications. There are few areas of active physics research that have not been effected by the AlexNet moment in some sense. While these applications have yet to produce a landmark discovery or new capability like the AlphaFold protein structure prediction model that was awarded the Nobel prize in chemisty in 2024, the breadth of these applications is a testatment to the power of

[2]Notable exlusions from this count are the Nobel prizes of Feynman, Schwinger, and Tomonaga for the QED theory which is a precursor of the Standard Model, and the Nobel prizes of Kajita and McDonald for the discovery of neutrino oscillations, which is notably not a part of the Standard Model.

[3]An excellent parallel to AlexNet is Charles Wilson's development of the cloud chamber, which was a tool developed by a meteorologist that became very useful for fundamental physics research and was awarded the Nobel a decade after it was developed.

deep learning for physics research.

In the remainder of the Introduction, I will outline the open problems in fundamental physics[4], and very briefly summarize the last 14 years of progress in machine learning and artificial intelligence. This review serves two purposes. The first is to give context to the very specific research detailed in this thesis. The second is to reflect on the fact that all open problems in particle physics are long-standing, and that the list is essentially unchanged since the 2012 discovery of the Higgs. These problems are extremely subtle and difficult, and many years of effort by very resourceful and intelligent scientists has so far proven insufficient to solve them. By contrast, the field of machine learning and artificial intelligence is completely unrecognizable to itself in 2012. The hope of this thesis, is that some of this dizzying progress can finally break the current impasse facing particle physics.

1.1 Open Problems in Fundamental Physics

The most pressing open problems in fundamental physics can be understood as either the need for a better understanding of Beyond the Standard Model (BSM) physics, or evidence of Beyond ΛCDM cosmology. BSM physics is known to exist through multiple experimental observations that can not be explained by the particle content and forces contained in the Standard Model. However the correct quantum field theory (or other type of theory consistent with the existing Standard Model) that explains these observations is unknown. ΛCDM cosmology accounts for all cosmological observations very well, with the exception of some significant tensions, namely the Hubble tension. My list of the open problems is as follows:

1. **Neutrino oscillations**: Neutrinos are massless under the Standard Model. However many iterations of neutrino experiments have verified that these particles "oscillate" between the three neutrino flavors as they propagate through space. This phenomenon

[4] I generally refer to the fields of particle physics and cosmology as "fundamental physics", given open questions do not necessarily sort neatly into the two sub-fields.

is easily explained if the neutrinos carry a mass that is small enough to remain currently undetected by experiments, but the mechanism that produces this mass is unknown. This fascinating topic is not covered at all in this thesis, but current and next generation neutrino experiments are making active use of deep learning.

2. **Dark matter**: A wide range of astrophysical and cosmological observations, including galaxy rotation curves [12], gravitational lensing, and the large-scale structure of the cosmic microwave background, indicate that roughly 27% of the energy budget of the universe is composed of a non-luminous, non-baryonic form of matter. The Standard Model contains no candidate particle that could account for this "dark matter". Many BSM dark matter candidate particles have been proposed, but none has ever been observed in a particle physics experiment. To the best of our knowledge, dark matter only interacts with the particles of the Standard Model through gravity.

3. **Matter–antimatter asymmetry**: The Standard Model predicts that the Big Bang should have produced equal amounts of matter and antimatter, yet the observable universe is composed overwhelmingly of matter. Introduction of this asymmetry requires breaking of the CP discrete symmetry. The known sources of CP violation in the Standard Model—the complex phase in the CKM quark mixing matrix and, to a lesser extent, the strong CP phase discussed below—are far too small to account for the observed asymmetry by many orders of magnitude [13]. A new source of CP violation beyond the Standard Model is therefore required, but remains unknown.

4. **Gravity**: The gravitational interactions are not present within the Standard Model. The general theory of relativity provides an extraodinarily accurate classical theory of gravity, but none of the attempts to quantize gravity and incoporate it with the other known forces has received experimental verification. At the energy scales probed by the LHC ($\mathcal{O}$(TeV)) gravitational effects are negligible, so this is not a direct experimental puzzle for collider physics, but our everyday experience of the gravitational force is

evidence of BSM physics.

5. **Origin of primordial density fluctuations**: The presence of the Baryon Acoustic Oscillation (BAO) peak in the CMB power spectrum is precisely predicted by ΛCDM cosmology, but requires the presence of primordial density perturbations to seed the oscillations. The mechanism that formed these perturbations in the early universe is not known, though inflation provides a compelling framework which requires the presence of a BSM field. Given there are competeing models and experimental evidence is scarce, this is less of a clean example of BSM physics than those listed above, but it is worth mentioning.

6. **Hubble tension**: The Hubble constant is a parameter of the ΛCDM model of cosmology that determines the expansion rate of the universe. This constant can be measured through two probes: the cosmic microwave background [14] and local distance-ladder observations [15]. However these two approaches yield values of the Hubble constant H_0 that disagree at the 4–5σ level. This tension could only be caused by an unaccounted for systematic effect and is under very active research. However if confirmed this would be a clear sign of beyond ΛCDM cosmology.

In addition to open problems related to experimental observations, there are many more open problems related to the theory of the Standard Model and its possible extensions. To make them more tractable I would group them into three categories. The first, and arguably the most salient at this moment in particle physics, are related to "fine-tunings" of the free parameters of the Standard Model and ΛCDM. Whether these fine-tunings are true problems or only an issue of philosophical priors is often debated, but I find them troubling enough to deserve mention here.

1. **The hierarchy problem**: The Higgs boson mass receives quantum corrections from every particle it couples to. Given its difficult to imagine a quantum theory of gravity

or generally any BSM model that doesn't couple to the Higgs directly or through loops, these quantum corrections should pull the Higgs mass up to the scale of the UV cutoff of the Standard Model. Instead the Higgs has a mass of 125 GeV. This requires either that the quantum corrections are cancelled by some new BSM physics near the electroweak scale, or that the scale of the corrections and the "bare mass" of the Higgs cancel nearly exactly despite their large size. This is the canonical example of a fine-tuning. Supersymmetry, compositeness, and extra dimensions are the most studied solutions to this problem, but none have been confirmed experimentally. The absence of any BSM signal at the LHC has sharpened this "naturalness" tension considerably.

2. **The strong CP problem**: Quantum chromodynamics (QCD) admits a CP-violating term in the Standard Model Lagrangian parameterized by $\bar{\theta}$. Measurements of the neutron electric dipole moment constrain $|\bar{\theta}| \lesssim 10^{-10}$, yet there is no symmetry in the Standard Model that forces this parameter to be small. This is another fine-tuning problem, though less relevant to LHC physics.

3. **The cosmological constant problem**: The observed energy density of the vacuum (dark energy) is roughly 10^{-47} GeV^4, yet naive estimates from quantum field theory predict a value $\sim 10^{120}$ times larger. This is arguably the worst fine-tuning problem in all of physics, and no compelling dynamical mechanism has been found to explain the observed small but non-zero value of the cosmological constant.

The second category consists of problems that are not fine-tunings, at least to the degree of the problems above, but are features of the Standard Model that in some sense "feel odd". Perosonally I don't discount such feelings as irrelevant as they have produced discoveries in the past[5].

[5]Dirac's prediction of the positron through aesthetic considerations about negative solutions to the Dirac equation is perhaps the most relevant one. The GIM mechanism is a particle physics example of how taking a fine-tuning seriously led to new discoveries (the charm quark).

1. **The flavor puzzle**: The Standard Model contains three generations of quarks and leptons that share identical gauge quantum numbers but differ enormously in mass, with Yukawa couplings spanning nearly five orders of magnitude from the electron to the top quark. No symmetry principle explains why there are exactly three generations, nor why the quark mixing angles (CKM matrix) are small while two of the neutrino mixing angles (PMNS matrix) are large. These hierarchies and mixing patterns are encoded in 13 of the 19 free parameters of the Standard Model with no motivation for their values. This is less of a fine-tuning and more of a structure that "feels odd". Personally I don't discount such feelings as irrelevant They have produced discoveries in the past.

2. **Accidental symmetries**: Several conservation laws of the Standard Model, baryon number B, and the individual lepton numbers L_e, L_μ, L_τ, are not imposed by the symmetries of the theory. Instead, they are accidents of the Standard Model gauge group, particle content, and renormalizability constraints. The fact that they are present and seem to hold experimentally is another of the "feels odd" problems.

3. **V-A structure of the weak interaction**: The weak interaction couples exclusively to left-handed fermions, a structure encoded in the $V - A$ (vector minus axial-vector) form of the weak current. Another way of stating this is that the weak interactions are maximally parity violating: the weak force would effect the dynamics of some particles but be completely turned off for their mirror image. While this is experimentally well established, there is no theoretical explanation within the Standard Model for why the weak force singles out left-handedness.

The final category is in my opinion the most important one. This is because it is indisputably experimentally and theoretically tracatable at this exact moment in particle physics, and has direct implications for how data from particle physics experiments are processed to hopefully make progress on the many problems quoted above. Furthermore it is the only set of open

questions that is directly relevant to the research presented in this thesis. It is problems relating to our ability to *calculate* within the Standard Model.

1. **Calculation of scattering amplitudes**: The Standard Model is predictive because it can be used to calculate scattering amplitudes that are tested every day in particle physics experiments. These amplitudes are almost always calculated at a fixed order in some perturbative expansion, and pushing known amplitudes to ever higher orders is an important open question in particle physics. In particular, improved modeling of Standard Model processes, whether as signal in measurements or background in searches, depends on making progress in this direction.

2. **Calculatibility of non-perturbative QCD**: Quantum chromodynamics is in principle a complete theory of the strong force, but the comparatively large value of the strong coupling constant α_s means perturbative expansions are only valid at high energies. At low energies, lattice QCD provides a systematic non-perturbative approach but is computationally expensive and limited in scope. This leaves us with few tools for modeling QCD dynamics at or below the QCD confinement scale of $\Lambda_{QCD} \sim 200$ MeV. This difficulty is one of the central bottlenecks in the interpretation of data from the LHC.

Experimental evidence is needed to guide theoretical progress on both of these problems. It is exactly that evidence that the research in this thesis provides at a uniquely large and flexible scale.

Finally, I will note that the best way to make progress on the many of these problems is unambiguously to find a new particle (necessarily one not in the Standard Model). Some historical data is relevant here. Figure 1.1 shows a timeline of the theoretical prediction or explanation (green) and experimental discovery (red) of all Standard Model particles. The rate of discoveries between 1965 and 1985, the "golden era" of particle physics, is clearly

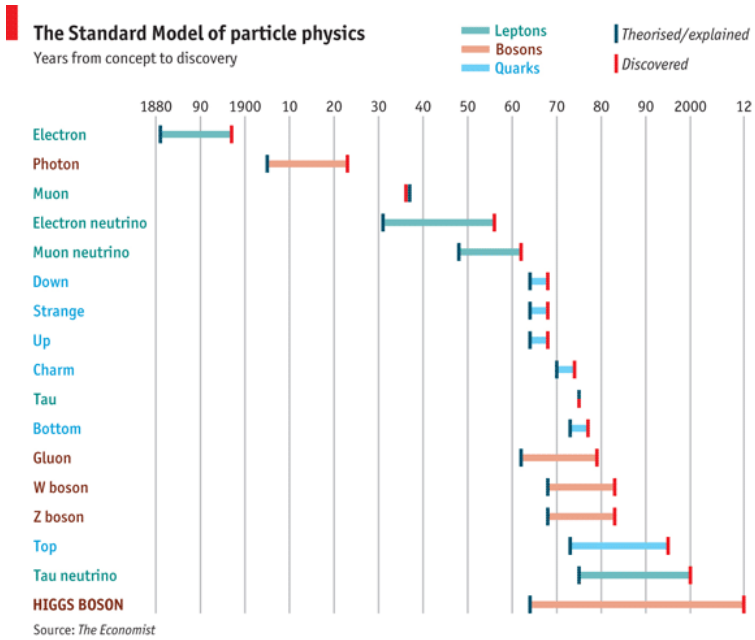

Figure 1.1: A timeline of the theortical prediction or explanation (green) and experimental discovery (red) of all Standard Model particles. **TODO: cite as fair use**

unique, as is the muon's experimental discovery before it was conceived of theoretically. However, it is also interesting to examine past droughts. Given the gap between the discovery of the electron and the photon can hardly be considered a time of drought for fundamental physics, the 20 year gap between the discovery of the muon in 1936 and the discovery of the electron neutrino in 1956 strikes me as the longest on record, and will remain so even if the next particle after the Higgs is not discovered for another couple years. Whatever geopolitical forces may have been at play in that time, particle physics research is clearly not always smooth sailing. However the next particle is soon to be overdue, and in my opinion prospects for finding it in the next few years are not fantastic. With that sobering fact, I'll contrast a field in which extraordinary progress has happened in the comparative blink of an eye.

1.2 The Deep Learning Revolution

Since the AlexNet moment in 2012, the field of machine learning and artificial intelligence has undergone rapid progress. The technology has matured from rather simple image classification models to AI systems that feel like science fiction in 14 years.

2013 - 2016: Following AlexNet, deeper networks with many millions of parameters provided even better performance on classification tasks [16, 17]. This was an early precursor of the scaling laws that are highly relevant to the current state-of-the-art. The particle physics community moved quickly: the first deep learning jet taggers appeared in 2014 [9, 10], only two years after AlexNet. This period also saw the development of generative neural networks, which produce samples additional samples from the underlying probability distribution of the training data. Generative Adversarial Networks (GANs) [18] and Variational Autoencoders (VAEs) [19] were early generative models that were picked up by the HEP community for generative tasks like fast calorimeter simulation and anomaly detection [20–23]. The most dramatic milestone of this period came in 2016 when AlphaGo [24] defeated Lee Sedol, the world champion Go player. Notably, RL has yet to find widespread application in high-energy physics, but its broader trajectory is relevant to the discussion of agentic AI in the conclusion.

2017: A next landmark was the introduction of the *attention mechanism* and the Transformer architecture in 2017 [25], which demonstrated state-of-the-art performance on machine translation tasks[6]. Attention's remarkable expressivity and the ability to parallelize the required computations on GPU resources meant Transformers replaced recurrent networks as the dominant sequence-processing architecture. The first HEP papers to use attention followed a few years later [26], and attention is now widely used by both ATLAS and CMS for

[6] It was also around this time that Google updated their Google Translate service to use neural machine translation. The development gathered enough media attention that I read about it as a first-year undergraduate. This touched off my interest in machine learning, and my decision to study physics was partially based on the fact that the CMS group was the only research group utilizing deep learning at my undergraduate institution.

hadronic object identification. Many of the neural networks discussed in this thesis consist of attention backbones.

2018 - 2022: Following the introduction of attention, the focus in ML/AI research was on scaling and generative modeling. (LLMs) trained on internet-scale text began to exhibit *emergent* capabilities, which are not present at smaller scales but appear abruptly as model size and training compute increase [27]. The CLIP model [28] demonstrated the ability of neural networks to process "multi-modal" data, for example by developing joint embeddings for text and data simultaneously. During this time the AI + HEP community was broadly focused on generative models. A lot of very rapid progress was driven by applying increasingly elaborate generative model architectures, from GANs to VAEs to normalizing flows [29], to diverse problems like jet generation [30], calorimeter simulation [21], unfolding [31–33], and anomaly detection [34, 35]. Then another landmark occured betwen 2021 and 2022 with the introduction of diffusion models [36, 37]. In the AI community, diffusion-based image generators like DALL-E 2, Stable Diffusion, and Midjourney made photorealistic image generation accessible to the general public for the first time. In HEP, diffusion models were quickly picked up and applied to the same set of generative tasks. Diffusion model based unfolding methods are a major topic in this thesis covered in Chapter 5. It is worth pausing to appreciate the distance traveled: neural networks had gone from classifying images to *generating* photorealistic images of arbitrary subjects in the space of 10 years.

2023: Another milestone occured in late 2023 with the introduction of ChatGPT. The introduction of a reinforcement learning (RL) based post-training [38] to LLMs transformed them from autocomplete engines into conversational assistants capable of writing code, reasoning through multi-step problems, and passing professional licensing examinations such as the bar exam. ChatGPT reached 100 million users in two months, faster than any consumer product in history, and the term "artificial intelligence" entered colloquial use.

2024 - 2026: Frontier LLMs reach scales of trillions of parameters trained on many terabytes

of text data. Simultaneously, advanced RL pipelines teach language models to use external tools, for example to edit files or search the web. This produces "agentic" AI systems that are capable of solving increasingly complicated and long-time horizon tasks autonomously. Massive labor market displacement due to these systems becomes a real concern, or more relavent to this thesis, some physics research papers are written almost entirely by AI systems under supervision from human experts [39, 40]. Whether or not this represents the future of scientific research, it is a rather remarkable development. I'll return to some opinions on autonomous physics research in the conclusion.

Before moving on, two points about the above deserve some attention. First, although the progress described above is remarkable but it has not been evenly distributed across all of machine learning. Many foundational problems remain as unsolved as they were in 2012. There is still no firm theoretical understanding of why stochastic gradient descent is so effective at training neural networks, and out-of-distribution generalization remains a challenge even on simple tasks. This bears some similarity to how fundamental questions in quantum mechanics were eventually set aside by the physics community because taking the formalism and running with it proved to be much more fruitful. The lesson is that fields move forward in unexpected directions, and that progress sometimes looks like re-focusing the question rather than stubbornly pursuing a solution. Second, the year 2022 strikes me as something of a turning point for the AI + HEP research program. Between 2012 and 2022, the main research directions of the broader AI community produced general-purpose data science tools that the HEP community could adapt relatively directly. Beating benchmarks on HEP tasks essentially required reading the latest NeurIPS proceedings and applying the new methods before other groups did the same. After 2022, the dominant direction of frontier AI research has shifted toward scaling, tool use, and the pursuit of general-purpose intelligence. The question of how pursuit of scaling maps onto fundamental physics is a fascinating one that I'll cover in Chapter 4, but generally speaking this direction does not map as cleanly onto the specific needs of fundamental physics. Without new tools coming

out of the frontier AI research community, the AI + HEP community has largely settled on Transformers and diffusion models as the core architectural toolkit. The task is now to use those tools to achieve concrete physics goals.

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Chapter 7

Conclusion

7.1 Machine Learning is Driving LHC Physics Progress

7.2 diHiggs: The Next Milestone

7.3 Agentic AI and the Transformation of Physics Research

7.4 Will AI Deliver the Next Particle Physics Discovery?

Bibliography

[1] ATLAS Collaboration. "Observation of a new particle in the search for the Standard Model Higgs boson with the ATLAS detector at the LHC". In: *Phys. Lett. B* 716 (2012), pp. 1–29. DOI: 10.1016/j.physletb.2012.08.020. arXiv: 1207.7214 [hep-ex].

[2] CMS Collaboration. "Observation of a New Boson at a Mass of 125 GeV with the CMS Experiment at the LHC". In: *Phys. Lett. B* 716 (2012), pp. 30–61. DOI: 10.1016/j.physletb.2012.08.021. arXiv: 1207.7235 [hep-ex].

[3] F. Englert and R. Brout. "Broken Symmetry and the Mass of Gauge Vector Mesons". In: *Phys. Rev. Lett.* 13 (9 Aug. 1964), pp. 321–323. DOI: 10.1103/PhysRevLett.13.321. URL: https://link.aps.org/doi/10.1103/PhysRevLett.13.321.

[4] Peter W. Higgs. "Broken Symmetries and the Masses of Gauge Bosons". In: *Phys. Rev. Lett.* 13 (16 Oct. 1964), pp. 508–509. DOI: 10.1103/PhysRevLett.13.508. URL: https://link.aps.org/doi/10.1103/PhysRevLett.13.508.

[5] G. S. Guralnik, C. R. Hagen, and T. W. B. Kibble. "Global Conservation Laws and Massless Particles". In: *Phys. Rev. Lett.* 13 (20 Nov. 1964), pp. 585–587. DOI: 10.1103/PhysRevLett.13.585. URL: https://link.aps.org/doi/10.1103/PhysRevLett.13.585.

[6] Steven Weinberg. "A Model of Leptons". In: *Phys. Rev. Lett.* 19 (21 Nov. 1967), pp. 1264–1266. DOI: 10.1103/PhysRevLett.19.1264. URL: https://link.aps.org/doi/10.1103/PhysRevLett.19.1264.

[7] G. 't Hooft and M. Veltman. "Regularization and renormalization of gauge fields". In: *Nuclear Physics B* 44.1 (1972), pp. 189–213. ISSN: 0550-3213. DOI: https://doi.org/10.1016/0550-3213(72)90279-9. URL: https://www.sciencedirect.com/science/article/pii/0550321372902799.

[8] Alex Krizhevsky, Ilya Sutskever, and Geoffrey E Hinton. "ImageNet classification with deep convolutional neural networks". In: *Advances in Neural Information Processing Systems*. Vol. 25. 2012, pp. 1097–1105.

[9] Pierre Baldi, Peter Sadowski, and Daniel Whiteson. "Searching for Exotic Particles in High-Energy Physics with Deep Learning". In: *Nature Commun.* 5 (2014), p. 4308. DOI: 10.1038/ncomms5308. arXiv: 1402.4735 [hep-ph].

[10] Josh Cogan et al. "Jet-Images: Computer Vision Inspired Techniques for Jet Tagging". In: *JHEP* 02 (2015), p. 118. DOI: 10.1007/JHEP02(2015)118. arXiv: 1407.5675 [hep-ph].

[11] HEP ML Community. *A Living Review of Machine Learning for Particle Physics*. URL: https://iml-wg.github.io/HEPML-LivingReview/.

[12] Vera C. Rubin and Jr. Ford W. Kent. "Rotation of the Andromeda Nebula from a Spectroscopic Survey of Emission Regions". In: *Astrophys. J.* 159 (1970), pp. 379–403. DOI: 10.1086/150317.

[13] A. D. Sakharov. "Violation of CP Invariance, C Asymmetry, and Baryon Asymmetry of the Universe". In: *Pisma Zh. Eksp. Teor. Fiz.* 5 (1967), pp. 32–35. DOI: 10.1070/PU1991v034n05ABEH002497.

[14] N. Aghanim et al. "Planck 2018 results. VI. Cosmological parameters". In: *Astron. Astrophys.* 641 (2020), A6. DOI: 10.1051/0004-6361/201833910. arXiv: 1807.06209 [astro-ph.CO].

[15] Adam G. Riess et al. "A Comprehensive Measurement of the Local Value of the Hubble Constant with 1 km/s/Mpc Uncertainty from the Hubble Space Telescope and the SH0ES Team". In: *Astrophys. J. Lett.* 934.1 (2022), p. L7. DOI: 10.3847/2041-8213/ac5c5b. arXiv: 2112.04510 [astro-ph.CO].

[16] Karen Simonyan and Andrew Zisserman. "Very Deep Convolutional Networks for Large-Scale Image Recognition". In: *International Conference on Learning Representations*. 2015. arXiv: 1409.1556 [cs.CV].

[17] Kaiming He et al. "Deep Residual Learning for Image Recognition". In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 2016, pp. 770–778. arXiv: 1512.03385 [cs.CV].

[18] Ian Goodfellow et al. "Generative Adversarial Nets". In: *Advances in Neural Information Processing Systems*. Vol. 27. 2014. arXiv: 1406.2661 [stat.ML].

[19] Diederik P. Kingma and Max Welling. "Auto-Encoding Variational Bayes". In: *International Conference on Learning Representations*. 2014. arXiv: 1312.6114 [stat.ML].

[20] Luke de Oliveira, Michela Paganini, and Benjamin Nachman. "Learning Particle Physics by Example: Location-Aware Generative Adversarial Networks for Physics Synthesis". In: *Comput. Softw. Big Sci.* 1.1 (2017), p. 4. DOI: 10.1007/s41781-017-0004-6. arXiv: 1701.05927 [stat.ML].

[21] Michela Paganini, Luke de Oliveira, and Benjamin Nachman. "Accelerating Science with Generative Adversarial Networks: An Application to 3D Particle Showers in Multilayer Calorimeters". In: *Phys. Rev. Lett.* 120.4 (2018), p. 042003. DOI: 10.1103/PhysRevLett.120.042003. arXiv: 1705.02355 [hep-ex].

[22] Michela Paganini, Luke de Oliveira, and Benjamin Nachman. "CaloGAN : Simulating 3D high energy particle showers in multilayer electromagnetic calorimeters with generative adversarial networks". In: *Phys. Rev. D* 97.1 (2018), p. 014021. DOI: 10.1103/PhysRevD.97.014021. arXiv: 1712.10321 [hep-ex].

[23] Olmo Cerri et al. "Variational Autoencoders for New Physics Mining at the Large Hadron Collider". In: *JHEP* 05 (2019), p. 036. DOI: 10.1007/JHEP05(2019)036. arXiv: 1811.10276 [hep-ex].

[24] David Silver et al. "Mastering the game of Go with deep neural networks and tree search". In: *Nature* 529.7587 (2016), pp. 484–489. DOI: 10.1038/nature16961.

[25] Ashish Vaswani et al. "Attention Is All You Need". In: *Advances in Neural Information Processing Systems*. Vol. 30. 2017. arXiv: 1706.03762 [cs.CL].

[26] Vinicius Mikuni and Florencia Canelli. "ABCNet: An attention-based method for particle tagging". In: *Eur. Phys. J. Plus* 135.6 (2020), p. 463. DOI: 10.1140/epjp/s13360-020-00497-3. arXiv: 2001.05311 [physics.data-an].

[27] Jason Wei et al. "Emergent Abilities of Large Language Models". In: *Transactions on Machine Learning Research* (2022). arXiv: 2206.07682 [cs.CL].

[28] Alec Radford et al. "Learning Transferable Visual Models From Natural Language Supervision". In: *Proceedings of the 38th International Conference on Machine Learning*. Vol. 139. Proceedings of Machine Learning Research. 2021, pp. 8748–8763. arXiv: 2103.00020 [cs.CV].

[29] Conor Durkan et al. "Neural Spline Flows". In: *Advances in Neural Information Processing Systems*. Vol. 32. 2019. arXiv: 1906.04032 [stat.ML].

[30] Anja Butter, Tilman Plehn, and Ramon Winterhalder. "How to GAN LHC Events". In: *SciPost Phys.* 7.6 (2019), p. 075. DOI: 10.21468/SciPostPhys.7.6.075. arXiv: 1907.03764 [hep-ph].

[31] Kaustuv Datta, Deepak Kar, and Debarati Roy. "Unfolding with Generative Adversarial Networks". In: (June 2018). arXiv: 1806.00433 [physics.data-an].

[32] Marco Bellagente et al. "Invertible Networks or Partons to Detector and Back Again". In: *SciPost Phys.* 9 (2020), p. 074. DOI: 10.21468/SciPostPhys.9.5.074. arXiv: 2006.06685 [hep-ph].

[33] Jessica N. Howard et al. "Learning to simulate high energy particle collisions from unlabeled data". In: *Sci. Rep.* 12 (2022), p. 7567. DOI: 10.1038/s41598-022-10966-7. arXiv: 2101.08944 [hep-ph].

[34] Benjamin Nachman and David Shih. "Anomaly Detection with Density Estimation". In: *Phys. Rev. D* 101 (2020), p. 075042. DOI: 10.1103/PhysRevD.101.075042. arXiv: 2001.04990 [hep-ph].

[35] Anna Hallin et al. "Classifying anomalies through outer density estimation". In: *Phys. Rev. D* 106.5 (2022), p. 055006. DOI: 10.1103/PhysRevD.106.055006. arXiv: 2109.00546 [hep-ph].

[36] Jonathan Ho, Ajay Jain, and Pieter Abbeel. "Denoising Diffusion Probabilistic Models". In: *Advances in Neural Information Processing Systems*. Vol. 33. 2020, pp. 6840–6851. arXiv: 2006.11239 [cs.LG].

[37] Yang Song et al. "Score-Based Generative Modeling through Stochastic Differential Equations". In: *International Conference on Learning Representations*. 2021. arXiv: 2011.13456 [cs.LG].

[38] Long Ouyang et al. "Training Language Models to Follow Instructions with Human Feedback". In: *Advances in Neural Information Processing Systems*. Vol. 35. 2022, pp. 27730–27744. arXiv: 2203.02155 [cs.CL].

[39] Matthew D. Schwartz. "Resummation of the C-Parameter Sudakov Shoulder Using Effective Field Theory". In: (Jan. 2026). arXiv: 2601.02484 [hep-ph].

[40] Alfredo Guevara et al. "Single-minus gluon tree amplitudes are nonzero". In: (Feb. 2026). arXiv: 2602.12176 [hep-th].

Appendix A

Supplementary Material

Appendix content goes here.